Hyeonjae Kim

Agency for Defense Development
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Research Interests

Programmable atomic systems for quantum simulation, quantum computing, and precision measurement

Education

Kyungpook National University (KNU) Daegu, Republic of Korea
Bachelor of Science in Physics Mar. 2020 – Feb. 2024
Overall GPA: 4.11/4.3 | Major GPA (71 credits): 4.19/4.3

Research Experience

Agency for Defense Development (ADD) Daejeon, Republic of Korea
Researcher (mandatory military service | PIs: Dr. Sangkyung Lee and Dr. Sin Hyuk Yim) Jun. 2024 – Present

- Developed an automated absorption spectroscopy system with spectral modeling to characterize atomic vapor cells and contributed to their fabrication [1]
- Built a zero-field optically pumped magnetometry platform to evaluate component-induced magnetic interference and provide design guidance for portable atomic sensors [4]
- Deployed a drone-mounted atomic magnetometer for magnetic anomaly detection and established a quantitative framework for interpreting the acquired data [2]

Korea Advanced Institute of Science and Technology (KAIST) Daejeon, Republic of Korea
Research Intern (MiliTECH Challenge)

- **Quantum and Nanobio Photonics Lab** (Advisor: Prof. Yong-Hoon Cho) Apr. 2023 – Dec. 2023
  Characterized $CsPbBr_3$ quantum dots using micro-photoluminescence spectroscopy
- **Aerospace Systems and Control Lab** (Advisor: Prof. Hyo-choong Bang) Apr. 2022 – Dec. 2022
  Designed a quadrotor with two hemispherical passive wheel shells and conducted mobility tests

Kyungpook National University (KNU) Daegu, Republic of Korea
Undergraduate Researcher

- **Novel Applied Nano Optics Lab** (Advisor: Prof. Junyeob Yeo) Jun. 2021 – Jun. 2022
  Optimized laser crystallization conditions for $BiVO_4$ photoelectrodes on flexible substrates
- **High Energy Physics Lab** (Advisor: Prof. Chang-Sung Moon) Jun. 2020 – Feb. 2021
  Developed momentum reconstruction algorithms for silicon detector simulations

Publications

† Corresponding author

[4] **H. Kim**, S. Lee†, S. H. Yim, T. Jeong, and Y. Lim, "Magnetic characterization of electronic components for portable atomic sensors using a zero-field optically pumped magnetometry platform," *Current Applied Physics*, under review (2026) [preprint]

[3] S. H. Hong, Y. Lim†, S. H. Yim, J. B. Nam, S. Lee, T. Jeong, and **H. Kim**, "Wire-sealed atomic gas cells for noble-gas nuclear magnetic resonance applications," *Review of Scientific Instruments*, **97**, 103203 (2026) [paper]

[2] **H. Kim**, S. H. Yim†, S. Lee, E.-S. Bang, S. R. Yeo, Y. J. Kim, T. Jeong, and Y. Lim, "Magnetic Anomaly Detection of an Undersea Tunnel Using a Drone-Mounted All-Optical Pulsed Atomic Magnetometer," *IEEE Sensors Journal* **26**, 30274–30281 (2026) [paper]

[1] **H. Kim**, T. Jeong, S. H. Hong, J. B. Nam, S. Lee, Y. Lim, and S. H. Yim†, "Aging test of atomic vapor cell with Al_2O_3 wall coating on cubic glass," *Applied Optics* **64**, 7932–7937 (2025) [paper]

Selected Presentations

† Corresponding author | Full list and abstracts [link]

H. Kim, S. Lee†, Y. Lim, S. H. Hong, T. Jeong, and S. H. Yim, "Probing magnetic noise from portable sensor components using a zero-field optically pumped magnetometer," *APS Division of Atomic, Molecular, and Optical Physics (DAMOP) Meeting*, Providence, RI, USA (2026); *poster presentation*

H. Kim†, "Experimental systems for spectroscopy and zero-field OPM with atomic vapor cells," *KISTI-SNU Joint Workshop*, Daejeon, ROK (2026); *invited talk*

H. Kim, T. Jeong, S. Lee, and S. H. Yim†, "Lifetime extension of rubidium vapor cells by Al_2O_3 coating," *APS Division of Atomic, Molecular, and Optical Physics (DAMOP) Meeting*, Portland, OR, USA (2025); *poster presentation*

Software Copyrights

H. Kim, S. Lee, T. Jeong, and S. H. Yim, "Buffer-Gas Pressure Estimation Software for Quantum Magnetometer Vapor Cells," *Korea Copyright Commission* C-2024-047156 (2024); *written in Python*

H. Kim, S. H. Yim, T. Jeong, and S. Lee, "Rubidium Vapor Cell Lifetime Assessment Software," *Korea Copyright Commission* C-2024-054414 (2024); *written in Python*

Skills

- Experimental: Atomic spectroscopy, magnetic field control, vacuum technology
- Programming: Python, Wolfram Language | Software: LabVIEW, SolidWorks, COMSOL Multiphysics, Origin

Honors and Awards

KFAS Study Abroad Workshop, *Korea Foundation for Advanced Studies* — Jul. 2026

Physics Alumni Association Award (Outstanding Graduate), *KNU Physics Alumni Association* — Feb. 2024

2nd Prize at the 2023 MiliTECH Challenge, *KAIST* — Dec. 2023
- Topic: Semiconductor quantum dot-optical fiber coupling for quantum cryptography

2nd Prize at the 2022 MiliTECH Challenge, *KAIST* — Dec. 2022
- Topic: Design of air-land quadrotor using hemisphere shell passive wheel

3rd Prize at the Academic Conference, *KNU College of Natural Sciences* — Dec. 2021
- Topic: Fabrication of a flexible photoelectrochemical cell using laser processing

Dean's List, *KNU College of Natural Sciences* — Nov. 2021, Apr. 2022, Apr. 2023

Scholarships

National Scholarship for Science and Engineering, *Korea Student Aid Foundation* — Mar. 2022 – Feb. 2024
- Full tuition and $8,000 in stipends

Hyoseok Scholarship, *KNU Alumni Association* — Apr. 2022
- Selected as the College of Natural Sciences representative; awarded $1,600

Academic Excellence Scholarship, *KNU Department of Physics* — Aug. 2020, Feb. 2021, Aug. 2021

Teaching Experience

Teaching Assistant, KNU Undergraduate Tutoring Program — Mar. 2023 – Dec. 2023
- Gave weekly two-hour lectures for Electromagnetism II (Spring) and Classical Mechanics II (Fall)

Tutor, KNU International Student Tutoring Program — Feb. 2022 – May 2022
- Provided weekly two-hour campus-life guidance and calculus tutoring; selected as an outstanding tutor